MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 1

# H1 Mathematics **8864**

**18 September 2013**

**3 hours**

Additional Materials: Writing paper
List of Formulae (MF 15)

**READ THESE INSTRUCTIONS FIRST**

Write your name and civics group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

**[Turn Over]**

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**Section A: Pure Mathematics [35 marks]**

**1** Given that $3(\ln x)^2 = 4 + 4\ln\left(\frac{1}{x}\right)$, use the substitution $u = \ln x$ to find the exact value of $x$. [5]

**2** **(a)** Sketch the graph of $y = \ln(3x+2)$. [2]

Hence, by adding a suitable graph, solve the inequality

$$5 - 2x > \ln(3x+2).$$ [2]

**(b)** Find the range of values of $p$ such that $2x^2 + 5px + 2 = 0$ has no real roots. [3]

**3** **(a)** Find $\int \frac{1}{2-3x} + \sqrt{3x+1}\ \mathrm{d}x$. [2]

**(b)** The curve $C$ has equation $y = \mathrm{e}^{2x} + 2$.

**(i)** Sketch the curve $C$. [2]

**(ii)** Find the exact area of the region bounded by the curve $C$, the line $x = 2$ and both axes. [3]

**4** The equation of a curve is $y = 2\ln(3-x)$.

**(i)** Find $\frac{\mathrm{d}y}{\mathrm{d}x}$. [1]

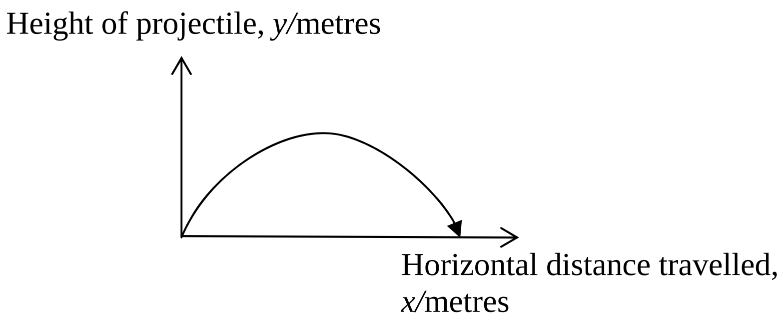
**(ii)** Hence find the equation of the normal to the curve at the point $D$ where $x = 1$. [3]

The normal to the curve at $D$ meets the $x$-axis and $y$-axis at $E$ and $F$ respectively.

**(iii)** Find the area of triangle $OEF$, where $O$ is the origin. [3]

**[Turn Over**

**5** The flight path of a projectile is shown in the diagram below.




The horizontal distance travelled by the projectile, $x$ metres, is related to the height of the projectile, $y$ metres, by the equation $y = 4\sqrt{10}x - x^2$.

**(i)** Using differentiation, find the maximum height that the projectile will reach. [5]

The height of the projectile at time $t$ seconds, is given by $y = 40t - 10t^2$.

**(ii)** Find the rate of change of the height of the projectile when $t = 0$. [1]

**(iii)** Hence find the rate of change of the horizontal distance travelled when $t = 0$. [3]

## Section B: Statistics [60 marks]

**6** A manufacturer produces mugs of 2 different sizes in the ratio $2:3$. He tests the quality of the mugs by taking a sample.

**(i)** Why does the manufacturer take a sample? [1]

**(ii)** He selects 20 mugs randomly from each size. State, with a reason, whether this will give a random sample of 40 mugs. [1]

**(iii)** Suggest and describe another sampling method that he can carry out in order to obtain a random sample of 40 mugs of different sizes. [3]

**7** A manufacturer claims that the Easy-bake bread mix he produces is packed in bags of 400 grams. It may be assumed that the mass of a randomly chosen bag of Easy-bake bread mix has a normal distribution with mean $\mu$ grams and standard deviation 5 grams. A sample of 50 bags is taken and the sample mean is found to be $\bar{x}$ grams. A hypothesis test conducted at the 5% level of significance yielded the result that the mean mass is not 400 grams.

**(i)** Find the set of values of $\bar{x}$. Give your answer to 1 decimal place. [6]

**(ii)** Hence state the conclusion if $\bar{x} = 399$. [1]

**(iii)** Explain the meaning of "at the 5% level of significance" in the context of the question. [1]

**(iv)** Explain whether it is necessary to assume that the mass of a randomly chosen bag of Easy-bake bread mix has a normal distribution. [1]

**8** The table shows the average cost of Certificate of Entitlement (COE), in thousand dollars, for Category A (cars below 1,600cc) for various years.

| Year, $x$ | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|---|---|---|
| Cost, $y$ | 40.1 | 29.2 | 27.5 | 18.8 | 12.9 | 14.7 | 14.3 | 5.6 |

**(i)** Give a sketch of the scatter diagram for the data, as shown on your calculator. [2]

**(ii)** Calculate the product moment correlation coefficient between $x$ and $y$. [1]

**(iii)** Find the equation for each of the regression lines $y$ on $x$ and $x$ on $y$. [2]

**(iv)** Use the regression line of $y$ on $x$ to estimate the year in which the average cost of COE drops to one thousand dollars. Comment on the reliability of this estimate. [3]

**(v)** Interpret, in context, the value of the gradient of the regression line of $y$ on $x$. [1]

**(vi)** Comment on whether a linear model would be appropriate, referring to the context of the question. [1]

**[Turn Over**

**9** **(a)** The pears sold at a supermarket are graded according to their masses. The masses of pears have a normal distribution. Pears with a mass less than 135 grams are graded as small, pears with a mass exceeding 200 grams are graded as large and the rest are graded as medium. Given that 25% of pears are small and 15% are large, find the mean and standard deviation of the distribution. [4]

**(b)** The masses, in kilograms, of apples and plums sold at the supermarket have independent normal distributions with means and standard deviations as shown in the following table:

| | Mean | Standard Deviation |
|---|---|---|
| Apples | 0.155 | 0.030 |
| Plums | 0.025 | 0.005 |

**(i)** Two apples and three plums are chosen at random. Find the probability that the total mass of the two apples and three plums is less than 0.36 kilogram. [3]

Apples are sold at \$4 per kilogram and plums are sold at \$12 per kilogram. Ms Woo buys three apples and three plums. Mr Tan buys ten plums.

**(ii)** Stating clearly the mean and variance of the distribution that you use, find the probability that Mr Tan pays more than Ms Woo by at least \$0.50. [4]

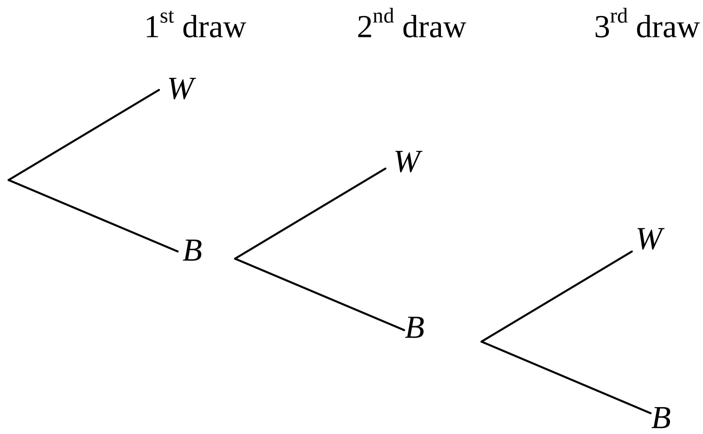
**10** **(a)** Two events $A$ and $B$ are such that $\mathrm{P}(A)=\dfrac{9}{20}$ , $\mathrm{P}(A\,|\,B)=\dfrac{6}{13}$ and $\mathrm{P}(A\cap B)=\dfrac{3}{10}$.

Find $\mathrm{P}(B)$ and $\mathrm{P}(A'\cap B')$. [5]

State, with a reason, whether $A$ and $B$ are independent. [1]

**(b)** A bag contains 5 black balls and 2 white balls. Elizabeth draws a ball repeatedly from the bag, at random and without replacement, until she gets a white ball.

**(i)** Copy the tree diagram below and fill in the probabilities for the first 3 draws where $W$ and $B$ indicate a white ball and a black ball drawn respectively. [1]




**(ii)** Find the probability that Elizabeth draws fewer than 4 balls. [2]

**(iii)** Find the probability that Elizabeth draws exactly 3 balls, given that she draws fewer than 4 balls. [3]

**[Turn Over**

**11** A random sample of $n$ people is chosen from a population and the number of people who have a particular gene A is denoted by $X$.

**(i)** State, in context, two assumptions needed for $X$ to be well modelled by a binomial distribution. [2]

Assume now that $X$ has the distribution $\mathrm{B}(n, 0.37)$.

**(ii)** Find the greatest value of $n$ such that the probability of having at most 10 people with gene A is more than 0.9. [2]

**(iii)** Given that $n = 10$, find $\mathrm{P}(3 < X \le 7)$. [2]

**(iv)** Given that $n = 200$, find the most probable number of people with gene A. [2]

**(v)** Given that $n = 200$, find $\mathrm{P}(X > 80)$ using a suitable approximation and explain why the approximation is appropriate in this case. [3]

**(vi)** Five random samples of 200 people each are chosen from the population. Find the probability that three out of the five samples have more than 80 people with gene A. [2]